Highlights

Background-Conditioned CFAR for Reliable Radar Target Detection in Nonstationary Sea Clutter

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- Background-conditioned CFAR calibration targets acquisition-level reliability.
- BC-DRCFAR reduces factor-2 false-alarm violations on synthetic sea clutter.
- Retrospective IPIX results improve acquisition-level false-alarm calibration.
- Robustness audits characterize temporal and domain reliability.

Background-Conditioned CFAR for Reliable Radar Target Detection in Nonstationary Sea Clutter

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Abstract

False-alarm control is central to radar target detection under nonstationary sea clutter. A detector can match the nominal rate after many trials are pooled while still producing acquisition-specific bursts of false alarms. We introduce BC-DRCFAR, a background-conditioned CFAR detector that uses same-time reference clutter to refine detection scores and set thresholds. BC-DRCFAR represents the background with distributional, temporal, tail-shape, and reference-series features and couples them to a scale-equivariant detection branch. Thresholds are calibrated directly to the declared false-alarm operating point. On the full-budget synthetic benchmark, BC-DRCFAR reduces the factor-2 false-alarm violation rate from 83.3% for GO-CFAR to 18.8%, lowers the mean absolute log10 false-alarm error from 1.070 to 0.174, and raises the detection probability at 0 dB from 0.740 to 0.835. On retrospective IPIX sea-clutter acquisitions, the feature-head calibration keeps the macro false-alarm probability close to the nominal level and improves acquisition-level calibration over scalar and GO baselines. Observed clutter background can thereby move CFAR beyond a local thresholding rule and make it a reliable calibration mechanism for nonstationary sea clutter.

Keywords: sea clutter, radar target detection, constant false alarm rate, probability of false alarm, background-conditioned calibration, IPIX radar

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data

1. Introduction

Reliable maritime radar target detection requires weak targets to be detected while false alarms stay at the declared operating point. Sea clutter makes this a calibration problem because the background return changes with sea state, grazing geometry, polarization, and acquisition conditions [1, 2, 3, 4]. A detector can show an acceptable average false-alarm probability after pooling many trials and still produce bursts of false alarms in individual acquisitions. For practical surveillance, this acquisition-level behaviour is the case operators encounter.

The declared false-alarm probability is more than a tuning parameter. It is the quantity that connects a detector to downstream tracking, display load, and operator response. When the empirical false-alarm rate drifts between acquisitions, the detector no longer provides the operating point promised by its threshold. This effect is especially important for small maritime targets, where raising the threshold to suppress clutter spikes can remove precisely the low signal-to-clutter returns that the detector is meant to preserve. Reliable sea-clutter detection therefore requires two capabilities at the same time: ranking target evidence and keeping the declared false-alarm rate meaningful in the background currently being observed.

Constant false-alarm rate (CFAR) detection remains the standard tool for this problem. Classical cell-averaging and greatest-of CFAR detectors estimate a local clutter level and set thresholds to meet a declared false-alarm probability [5, 6, 7]. Their simplicity and clear operating-point interpretation make them strong baselines. In nonstationary sea clutter, heavy-tailed and mismatched reference cells directly disrupt threshold calibration. This turns false-alarm control from a local power-estimation problem into a background-conditioned calibration problem.

The key technical difficulty is that the reference window has two roles. It provides the samples from which the clutter level is estimated, and it also reveals the condition under which the threshold will be used. Classical CFAR mostly compresses this window into a scalar summary, such as an average, an order statistic, or a greatest-of estimate. That summary is effective when the background is close to the assumed model. In measured sea clutter, the same window can also contain information about tail weight, temporal persistence, range-bin mismatch, and polarization-dependent clutter structure. Treating that information as a calibration signal is the central move in this paper.

Recent work has improved radar detection in heterogeneous and non-Gaussian sea clutter through adaptive CFAR variants, statistical clutter modelling, feature-based detection, and learning-based detectors [8, 9, 10, 11, 12]. These methods have advanced robustness in important regimes. A gap remains at the calibration stage: the same background structure used for detection should also inform the declared false-alarm operating point. BC-DRCFAR develops this idea by conditioning calibration on the observed clutter structure that drives acquisition-specific false alarms.

We propose BC-DRCFAR as a background-conditioned CFAR framework for reliable radar target detection in nonstationary sea clutter. The method uses same-time reference clutter to condition both score refinement and threshold adjustment. It represents the background using distributional, temporal, tail-shape, and reference-series features, then couples these features to a scale-equivariant detection branch. The resulting detector keeps the CFAR operating-point interpretation while adding a learned calibration mechanism driven by the clutter background observed at detection time.

The main contributions are as follows.

- We formulate sea-clutter false-alarm control as an acquisition-level calibration problem, exposing failures that can be hidden by pooled P_{fa} summaries.
- We propose a background-conditioned CFAR architecture that jointly refines detection scores and calibrates thresholds from same-time reference clutter.
- We combine distributional, temporal, tail-shape, and reference-series background descriptors with a scale-equivariant detection branch, giving the model direct access to the clutter structure that governs false-alarm instability.
- We validate BC-DRCFAR on a full-budget synthetic benchmark and retrospective McMaster IPIX sea-clutter acquisitions, showing improved false-alarm calibration over GO-CFAR and scalar calibration baselines.

2. Related work

2.1. CFAR detection in heterogeneous sea clutter

CFAR detection is the standard radar framework for converting a desired false-alarm probability into an adaptive threshold. Classical CA-CFAR, GO-CFAR, SO-CFAR, and OS-CFAR variants differ mainly in how the reference window is summarized and how interference or clutter edges are handled [6, 5, 13]. This design gives CFAR detectors a clear operational interpretation, which is why they remain strong baselines in sea-clutter studies.

Maritime clutter often departs from the homogeneous background assumptions that make simple reference-window estimates reliable. High-resolution sea clutter can be heavy-tailed, non-Gaussian, and heterogeneous across adjacent range cells and acquisitions. Much of the CFAR literature addresses this by adopting more suitable clutter models or more robust reference-cell processing. Examples include heavy-tailed texture models, K-distributed clutter modelling, heterogeneous-clutter CFAR design, and environment-adaptive thresholding [14, 15, 8, 16, 10, 9, 17, 18, 19]. These methods establish the value of adapting CFAR to clutter statistics. BC-DRCFAR extends this line of work by treating the background as a conditioning variable for calibration, so that threshold adjustment is driven by the observed clutter structure at detection time.

The difference becomes clear at the acquisition level. Classical CFAR analysis usually emphasizes the threshold factor implied by a clutter model and a target false-alarm probability. In measured sea clutter, the same nominal setting can produce different false-alarm behaviour across acquisitions because the reference cells carry different tail, correlation, and mismatch structure. BC-DRCFAR uses this acquisition-specific structure directly. It preserves the CFAR idea of declaring an operating point while allowing the calibration rule to respond to the observed reference background.

This acquisition-level view also changes how methods should be compared. A detector can match the nominal P_{fa} after all clutter cells are pooled and still be unreliable in the individual acquisition that generated the alarms. For that reason, the relevant comparison must include the threshold formula, the pooled false-alarm rate, and the stability of the declared operating point across acquisitions. BC-DRCFAR is built for this comparison: the input is the same reference background used by CFAR, and the output remains a thresholded decision at a declared P_{fa}.

2.2. Feature-based and statistical sea-clutter detection

Another line of work improves maritime target detection by describing sea clutter with richer statistics before applying a detector. Filtering-then-CFAR methods suppress clutter in the spatio-temporal or frequency-wavenumber domain and then design a detector for the filtered background [20, 21]. Other approaches model nonlinear feature dependence or exploit multi-channel observations. Copula-based CFAR methods, for example, use feature dependence to derive detection statistics and false-alarm probabilities for multi-feature or dual-polarization detection [22, 23, 24].

These studies show that background structure carries information beyond a single amplitude estimate. BC-DRCFAR uses that structure for calibration as well as detection. Its distributional, temporal, tail-shape, and reference-series descriptors condition both score refinement and threshold selection.

This also separates the proposed detector from feature-only decision rules. Feature-based detectors often improve the separation between target and clutter returns, but the link between a multidimensional feature space and a declared P_{fa} can become indirect. BC-DRCFAR keeps that link explicit by using background descriptors to set the threshold itself. The features are not only evidence for the target class; they are evidence about the clutter state under which the threshold must operate.

The distinction is important for sea clutter because the same feature can have different decision meaning under different backgrounds. A high-amplitude return inside a calm, weak-tailed reference window is different from a high-amplitude return inside a spiky, correlated, or polarization-shifted background. A feature-only detector tends to ask whether the CUT is target-like. A background-conditioned CFAR detector also asks what threshold is justified by the clutter condition around that CUT. This makes feature extraction part of the false-alarm calibration mechanism and ties the features directly to the operating point [25, 26, 27].

2.3. Learning-based radar target detection

Learning-based radar detectors have been introduced to capture target–clutter differences that are difficult to express with fixed statistical models. Recent methods combine reconstruction, multi-domain features, neural networks, or geometric representations to improve detection under complex sea clutter [28, 29, 30, 31, 32, 33, 34]. These approaches can absorb heterogeneous inputs and learn discriminative representations from measured data.

For radar deployment, the decision threshold must correspond to a reliable false-alarm operating point. BC-DRCFAR places declared P_{fa} calibration at the center of the method. It keeps the CFAR operating-point language and uses a learned background-conditioned mechanism to improve acquisition-level false-alarm reliability in nonstationary sea clutter.

The contribution is complementary to learning-based representation methods. A detector may learn a strong score and still require a calibration rule that remains meaningful from acquisition to acquisition. BC-DRCFAR makes this rule part of the model. The score branch orders target evidence, and the threshold branch maps the declared operating point to the local clutter condition. This design gives the learned detector a CFAR-style operating interface instead of a free classification threshold.

This paper therefore connects two lines of radar signal processing that are often treated separately. Classical CFAR supplies a disciplined operating interface through P_{fa}, while modern feature learning supplies a way to represent clutter structure beyond a scalar reference estimate. BC-DRCFAR keeps the operating interface and uses the learned representation only where it is most useful: calibrating the threshold for the observed background.

3. Method

3.1. Detection problem and CFAR operating point

For each cell under test (CUT), the detector receives a complex radar sample and a set of same-time reference cells from the same acquisition. Let x denote the CUT and let $R = \{r_i\}_{i=1}^{K}$ denote the reference set. The task is target detection at a declared false-alarm operating point α.

We write the detector as a score function and a threshold function. The score $s(x, R)$ measures target evidence in the CUT under its local clutter context. The threshold $\tau(R; \alpha)$ maps the same context and the declared false-alarm probability to a decision level. A detection is declared by

$$d(x, R; \alpha) = \mathbb{1}\{s(x, R) > \tau(R; \alpha)\}. \tag{1}$$

This form preserves the CFAR operating-point language while allowing the threshold to depend on the observed background structure.

The formulation separates two quantities that are often coupled in practice: target evidence and false-alarm calibration. The score $s(x, R)$ ranks how target-like the CUT appears under its local context. The threshold $\tau(R; \alpha)$

expresses the decision level required by the declared operating point in that same context. This separation lets BC-DRCFAR improve the ordering of candidate cells without losing the CFAR interpretation of the final binary decision.

3.2. Overview of BC-DRCFAR

BC-DRCFAR separates background description, CUT scoring, and threshold calibration. The background branch extracts descriptors from the same-time reference clutter. The CUT branch computes target evidence for the test cell and conditions that evidence on the background representation. The threshold branch then maps the same background representation and target α to the decision level used in Eq. (1).

After training, the inference path is fixed. Given (x, R, α), the detector forms a background representation from R, refines the CUT score under that background, evaluates the calibrated threshold, and compares the two quantities. The background enters the detector as an explicit calibration variable.

The architecture is built around the same information available to a CFAR detector at test time. Calibration factors and feature-head parameters are fixed before retrospective scoring. Acquisition-specific behaviour is then carried by the measured reference background. This fixed inference path makes the IPIX evaluation a direct test of calibration transfer.

The resulting pipeline has a simple operational interpretation. The reference cells are first converted into a background representation. The CUT is then scored relative to that representation. Finally, the declared operating point α is converted into a threshold that is specific to the same background. The method therefore keeps the CFAR decision form and upgrades the calibration step from a scalar threshold correction to a background-conditioned threshold function.

Figure 1 summarizes this data flow and fixes the role of each module before the module-level description. The background encoder, score branch, threshold branch, and final score-threshold comparison together form a single CFAR-style decision rule rather than a detached scoring network.

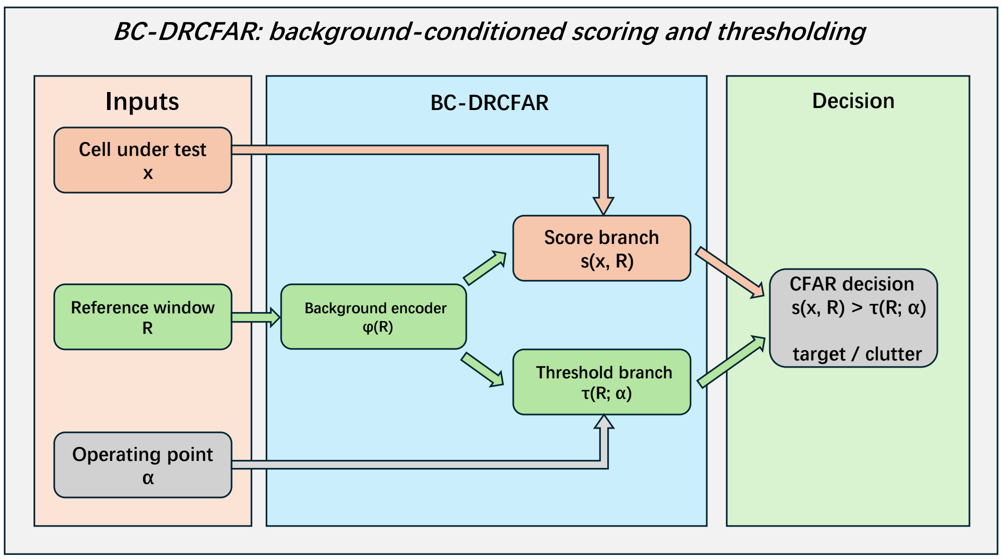


Figure 1: Overview of BC-DRCFAR. Same-time reference clutter is represented by distributional, temporal, tail-shape, and reference-series descriptors. The background representation conditions both the CUT score and the threshold used for the declared false-alarm operating point.

3.3. Background representation

The background branch summarizes the same-time reference clutter with four descriptor groups. Distributional descriptors record the amplitude level and spread of the reference cells. Temporal descriptors record dependence across the available slow-time samples. Tail-shape descriptors record heavy-tailed and spiky behaviour in high-resolution sea clutter. Reference-series descriptors retain information from the ordered reference sequence, so the background is not collapsed to a single power estimate.

Each descriptor group has a distinct calibration role. Distributional descriptors anchor the scale of the local clutter. Temporal descriptors capture slow-time persistence, which changes how isolated high returns should be interpreted. Tail-shape descriptors summarize the tendency of sea clutter to produce spikes that resemble target returns. Reference-series descriptors keep the ordering and local contrast information that is lost when the reference window is reduced to one average or order statistic.

Let $\phi(R)$ denote the concatenated background descriptor vector. BC-DRCFAR uses $\phi(R)$ as the conditioning variable for both score refinement and threshold calibration:

$$z_R = g_\theta(\phi(R)), \tag{2}$$

where g_θ is the background encoder and z_R is the fused clutter representation. This vector carries the background factors used later for score refinement and threshold calibration.

The same z_R is shared by the score and threshold branches. Sharing the background representation makes the two branches respond to the same observed clutter state. When the background is heavy-tailed, shifted, or temporally correlated, both the target-evidence score and the decision threshold are conditioned on that state instead of being calibrated from a pooled average.

The four descriptor groups are intentionally redundant at the level of raw information but distinct at the level of calibration role. Distributional features supply the local scale and spread that classical CFAR estimates with a single statistic. Temporal features describe slow-time dependence, which affects the chance that a high return is part of a persistent clutter process. Tail-shape features capture spikiness and asymmetric high-end behaviour, the main source of false alarms in heavy-tailed clutter. Reference-series features preserve ordered local structure across nearby non-target cells, giving the threshold branch access to mismatch patterns that are lost in scalar averaging.

3.4. Scale-equivariant score refinement

The CUT branch computes a base target-evidence score from the test cell and its reference context. Radar amplitudes can vary substantially across acquisitions, so the score path is designed to preserve the decision structure under common amplitude rescaling. In practice, the CUT branch operates on normalized CUT and reference statistics and is coupled to z_R before the final score is produced:

$$s(x, R) = f_\theta(x, R, z_R). \tag{3}$$

This branch provides the target-evidence ordering used by the final CFAR decision.

Scale equivariance is used to keep the score meaningful across acquisitions with different amplitude ranges. If both the CUT and its reference background are scaled by a common factor, the normalized statistics preserve the relative target evidence seen by the branch. The final threshold can then handle the declared false-alarm operating point without forcing the score branch to learn a new absolute scale for every acquisition [35].

3.5. Background-conditioned threshold calibration

The threshold branch is the calibration core of BC-DRCFAR. For a declared false-alarm probability α, the branch predicts a threshold from the background representation:

$$\tau(R; \alpha) = h_\theta(z_R, \alpha). \tag{4}$$

The threshold function is anchored to monotone P_{fa} operating points. Lower target false-alarm probabilities correspond to stricter thresholds, and the learned correction adjusts the threshold according to the observed clutter structure. The score path orders candidate CUTs, while the threshold path maps the declared operating point to the acquisition-specific decision level.

The calibration branch is anchored by analytic CFAR-style summaries before the background-conditioned correction is applied. The anchor set includes summaries based on CA power, trimmed power, Rayleigh median behaviour, Weibull log-moment behaviour, upper order statistics, and robust high-quantile reference levels. These anchors give the threshold branch a physically interpretable starting point across common clutter regimes. The learned residual then adjusts the threshold multiplicatively as a function of z_R and α, keeping the correction tied to the declared false-alarm scale and separate from an arbitrary classification score.

This design gives the method a direct link to CFAR practice. The user still chooses an operating point α, and the detector still returns a binary decision by comparing a score with a threshold. The learned component changes how the threshold is calibrated for the observed background. In this sense, BC-DRCFAR keeps the operational interface of CFAR while replacing scalar threshold adjustment with background-conditioned calibration.

3.6. Staged training and inference

Training is staged to keep detection and calibration identifiable. Stage A learns the score path on the development split. The score definition is then frozen. Stage B trains residual CUT conditioning and the threshold-calibration branch with the score path fixed. At test time, the frozen model receives (x, R, α), computes $\phi(R)$ and z_R, evaluates $s(x, R)$ and $\tau(R; \alpha)$, and returns the decision from Eq. (1).

The staged procedure gives each training phase a clear role. Stage A learns a stable ordering of CUT evidence. Stage B then learns how much the decision level should move for the declared P_{fa} under the observed background. Freezing the score before threshold calibration prevents the model from hiding calibration changes inside the score scale. The final detector is therefore a fixed decision rule at evaluation time.

Model selection follows the reliability objective used in the paper. Candidate checkpoints are compared first by acquisition-level factor-2 violation rate and log-scale false-alarm error, then by matched-operating-point detection probability, parameter count, and latency. This ordering makes calibration stability the primary selection criterion and treats detection probability as a paired requirement at the same declared P_{fa}. The parameter budget is capped at 500,000 trainable parameters, so the reported gains are tied to the calibration design and a compact model class.

3.7. Reliability metrics

The primary reliability endpoint is acquisition-level false-alarm calibration. For each acquisition or series, the empirical false-alarm probability $\widehat{P}_{\text{fa}}$ is compared with the declared target α. We report the factor-2 violation event

$$\widehat{P}_{\text{fa}} < \frac{\alpha}{2} \quad \text{or} \quad \widehat{P}_{\text{fa}} > 2\alpha. \tag{5}$$

We also report the mean absolute log10 false-alarm error and detection probability P_{d} at the specified signal-to-clutter ratio. These metrics separate pooled calibration, acquisition-level calibration, and detection probability.

The factor-2 event is used because it is directly interpretable at the operating-point level. A detector with good pooled P_{fa} can still violate the declared rate in many individual acquisitions. Mean absolute log10-P_{fa} error measures the size of these deviations on the natural scale for false-alarm probabilities. Reporting P_{d} alongside these calibration metrics keeps the detection trade-off visible.

4. Experiments

4.1. Experimental design

The experiments follow the reliability problem addressed by BC-DRCFAR. We first use a controlled nonstationary sea-clutter benchmark to measure acquisition-level false-alarm calibration. We then move to public retrospective IPIX acquisitions and keep the calibration artifacts frozen. Ablations and reliability audits separate the effect of background conditioning from scalar calibration and standard GO-CFAR thresholding.

All reported evaluations use locked data splits and fixed operating points. The main target false-alarm probability is $P_{\mathrm{fa}} = 0.01$. Two stricter operating points, $P_{\mathrm{fa}} = 0.003$ and $P_{\mathrm{fa}} = 0.001$, are also reported on the synthetic benchmark. The main comparison is made at the acquisition level, where false-alarm bursts directly determine operational reliability.

The evaluation protocol is deliberately tied to the operating point and to acquisition-level reliability. For each method, the declared P_{fa} is fixed before scoring. The observed false-alarm probability is then measured both after pooling all cells and within each acquisition. This layout exposes the practical calibration question: a radar detector must keep the declared operating point in the series being processed as well as in the pooled summary.

4.2. Synthetic nonstationary sea-clutter benchmark

The synthetic benchmark contains seven clutter scenarios: candidate-family, gamma-shape shift, inverse-gamma G^0, correlated, contaminated, mixture, and state-switching clutter. Each scenario is evaluated across four severity levels, three slow-time lengths, three reference-cell counts, and three target P_{fa} values. The nominal target $P_{\mathrm{fa}} = 0.01$ is the primary endpoint; 0.003 and 0.001 evaluate stricter false-alarm operation. Splits are frozen before evaluation.

The synthetic protocol fixes 36 cells for each scenario-severity stratum and uses deterministic seeds for cell identities, clutter parameters, slow-time sequences, and target injections. Each stratum is split into 18 development cells, 6 calibration cells, and 12 locked-test cells. Target returns are injected at specified signal-to-clutter ratios, while the clutter-only reference portion remains tied to the scenario and acquisition cell. This design makes the calibration test sensitive to changes in clutter mechanism, reference-cell count, and slow-time length while keeping the evaluation unit fixed.

Because the clutter family, severity, slow-time length, reference-cell count, and target operating point are fixed by design, this benchmark gives a direct readout of calibration under explicit nonstationarity.

The synthetic benchmark provides the cleanest separation between clutter condition and detector behaviour. Candidate-family and gamma-shape settings test model mismatch in the clutter law. Inverse-gamma G^0 and contaminated clutter stress heavy tails and outliers. Correlated and state-switching settings test slow-time dependence and nonstationary transitions. Mixture clutter combines several of these effects in one series. This range lets the evaluation measure whether the calibration rule follows the observed background structure across distinct nonstationary mechanisms. Table 1 summarizes the benchmark factors and the frozen split used for the controlled calibration test.

Table 1: Synthetic benchmark design. The controlled benchmark varies clutter mechanism, severity, slow-time support, reference support, and target false-alarm operating point under frozen splits.

Design element	Setting
Clutter scenarios	Candidate-family, gamma-shape shift, inverse-gamma G^0, correlated, contaminated, mixture, and state-switching clutter
Severity levels	Four levels per clutter scenario
Slow-time lengths	Three lengths per setting
Reference-cell counts	Three reference-cell counts per setting
Target operating points	$P_{\mathrm{fa}} = 0.01$ as the primary endpoint, with 0.003 and 0.001 as stricter operating points
Cell allocation	36 cells per scenario-severity stratum: 18 development, 6 calibration, and 12 locked-test cells
Controlled variables	Cell identities, clutter parameters, slow-time sequences, target injections, and data splits are fixed before evaluation
Primary readout	Acquisition-level factor-2 false-alarm violation, log10-P_{fa} error, and P_{d} at the declared operating point

4.3. Retrospective IPIX evaluation

The retrospective real-data experiment uses public McMaster IPIX stare acquisitions. Five acquisitions are used only to estimate calibration factors or feature-head parameters. Nine acquisition-disjoint retrospective acquisitions are then scored with frozen artifacts [36, 3].

The IPIX protocol uses a target P_{fa} of 0.01, 128-pulse nonoverlapping slow-time blocks, and at most 24 same-time reference cells. Four polarizations, HH, VV, HV, and VH, are retained when available. For each acquisition and polarization, one common mean-scale-phase transform is fitted from official non-target bins and then applied consistently; no per-range-bin rescaling is introduced. The CUT is one 128-pulse block from one range bin, and the reference set is formed from nearest official non-target bins at the same time, capped at 24 cells with variable K kept explicit.

The IPIX cohort provides the main measured-data evaluation. Its acquisition-disjoint split keeps development calibration separate from retrospective scoring.

The retrospective design matches the way a calibrated detector is used after deployment. Calibration factors and feature-head parameters are estimated on the development acquisitions. The retrospective acquisitions are then evaluated with the frozen artifacts and the declared operating point. This split makes the real-data result a direct acquisition-transfer experiment under measured sea clutter.

4.4. Baselines and model variants

The primary CFAR baseline is GO-CFAR, which provides a strong classical reference for clutter-edge handling and declared false-alarm operation. The calibration baselines are scalar calibration and low-rank calibration variants. The main BC-DRCFAR model is the feature-head variant, which conditions the threshold on the background descriptor representation. Ablation variants remove or modify background feature groups to quantify each descriptor family's contribution to the final calibration behaviour.

The real-data comparator set also includes CA-CFAR, OS-CFAR, and trimmed-mean CFAR in the protocol, with GO-CFAR used as the principal classical reference in the manuscript tables. Scalar calibration estimates one multiplicative factor from the development acquisitions and applies it unchanged to retrospective acquisitions. The low-rank variant gives the calibration rule additional degrees of freedom without using the full background feature head. These comparisons separate three possibilities: a classical

CFAR threshold may be enough, a global threshold shift may be enough, or acquisition-conditioned features may be needed for stable false-alarm control.

All baselines and variants are evaluated under the same target P_{fa} and the same acquisition splits. Data access and operating-point selection remain fixed across the comparison.

The baseline set separates three sources of performance. GO-CFAR tests a classical CFAR rule with a strong clutter-edge interpretation. Scalar calibration tests whether a global threshold correction is enough once the score is fixed. Low-rank calibration tests a more flexible correction without the full feature-head conditioning. The feature-head variant tests the central claim of BC-DRCFAR: acquisition-level false-alarm reliability improves when the threshold is conditioned on the observed background descriptors.

4.5. Evaluation metrics

The main metrics are pooled P_{fa}, acquisition-level P_{fa}, factor-2 false-alarm violation rate, mean absolute log10-P_{fa} error, and detection probability P_{d}. Pooled P_{fa} measures average calibration over all evaluated cells. The acquisition-level metrics measure preservation of the declared operating point within each acquisition. P_{d} is reported at the specified signal-to-clutter ratio to show the detection trade-off associated with calibration.

These metrics are read together. Pooled P_{fa} confirms that a method is calibrated on average. The factor-2 metric counts acquisitions whose observed false-alarm rate moves outside a practical tolerance band. The log-scale error measures how far those deviations move from the declared value. P_{d} then shows whether the calibration gain is obtained while preserving target detection performance.

For the real-data cohort, the acquisition is the primary statistical unit. Polarization, range-bin, and slow-time blocks are nested observations within an acquisition. This reporting choice matches the operational question: a detector is useful when the acquisition being processed respects the declared false-alarm rate. It also keeps the comparison between scalar calibration and feature-head calibration focused on transfer across measured acquisitions with blocks nested inside each file.

4.6. External reliability audits

Reliability audits cover temporal, source, and external-domain behaviour. The long-horizon IPIX holdout measures temporal drift under a fixed evaluation protocol. Fold-level source audits identify real source difficulty. Birmingham 626 is used for raw-data accessibility and MAT/HDF5 chunk recovery; the latest audit recovered three HDF5 raw-data chunk nodes, 171 indexed chunk records, and 148 successfully decoded chunk payloads from the cached prefix. St Andrews and related external-domain checks extend the transfer evaluation beyond the main IPIX setting [37, 38, 39].

5. Results

The results follow the evaluation route defined above. The synthetic benchmark tests calibration when the clutter source and operating point are controlled. The IPIX evaluation then applies frozen calibration artifacts to acquisition-disjoint measured data. Ablations isolate the background information used by the feature head, and temporal, source, and external-domain audits close the reliability analysis.

5.1. Controlled benchmark: BC-DRCFAR reduces false-alarm excursions

On the full-budget synthetic benchmark, BC-DRCFAR improved acquisition-level calibration relative to GO-CFAR. At the primary operating point, the pooled P_{fa} of BC-DRCFAR was 0.01012, close to the nominal value of 0.01. The acquisition-level factor-2 violation rate fell from 83.3% for GO-CFAR to 18.8%. The mean absolute log10-P_{fa} error decreased from 1.070 to 0.174, and P_{d} at 0 dB increased from 0.740 to 0.835. The controlled benchmark shows that background-conditioned thresholds reduce acquisition-level false-alarm excursions.

Figure 2 gives the visual version of the synthetic result. It brings together the three quantities that define the main claim: acquisition-level factor-2 false-alarm violations, log-scale false-alarm error, and detection probability at the matched operating point.

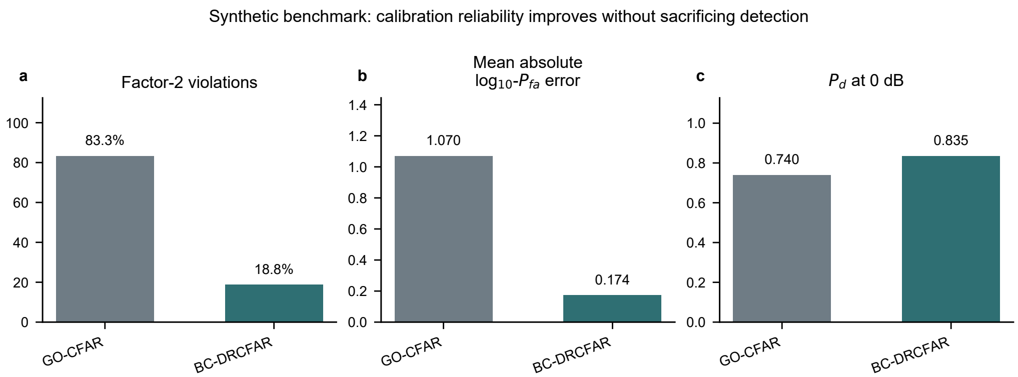


Figure 2: Synthetic calibration result at the nominal operating point. BC-DRCFAR reduces acquisition-level false-alarm excursions and log-scale calibration error while preserving target detection at 0 dB.

The main improvement is acquisition-level calibration reliability. The pooled P_{fa} confirms that the declared operating point is matched on average, while the factor-2 metric shows how often individual conditions remain close to that point. The drop from 83.3% to 18.8% means that BC-DRCFAR suppresses the acquisition-specific false-alarm bursts that are hidden by pooled summaries. The simultaneous increase in P_{d} at 0 dB shows that the calibration gain is obtained while retaining weak-target detection performance.

The synthetic result is the strongest evidence for the calibration mechanism because the sources of nonstationarity are controlled. Each clutter family changes a different part of the background: tail weight, correlation, mixture composition, contamination, or state transition. A scalar threshold factor has to absorb these changes through one number. BC-DRCFAR instead receives a descriptor vector from the same reference background and can adjust the threshold according to the mechanism expressed in that vector. The improvement in both log-P_{fa} error and factor-2 violations is therefore consistent with the intended role of the background branch. The main numerical comparison is reported in Table 2.

Table 2: Primary synthetic benchmark summary at the nominal operating point.

Method	Pooled P_{fa}	Factor-2 violation	MAE log10-P_{fa}	P_{d} at 0 dB
GO-CFAR	–	0.833	1.070	0.740
BC-DRCFAR	0.01012	0.188	0.174	0.835

5.2. Retrospective IPIX: feature-head calibration improves reliability

The retrospective IPIX cohort showed the same calibration pattern. With frozen calibration artifacts, the feature head produced a macro P_{fa} of 0.009998. Its acquisition-level absolute log-P_{fa} error was 0.199, compared with 0.213 for scalar calibration and 0.299 for GO-CFAR. The feature head satisfied the acquisition factor-2 condition in 7 of 9 acquisitions. The primary detection probability was 0.282, close to the GO-CFAR value of 0.286. The measured-data gain lay mainly in false-alarm calibration, with little change in the detection trade-off.

Figure 3 places the retrospective IPIX result at the acquisition level. The panels separate acquisition-level false-alarm error, factor-2 status, and primary-bin P_{d}, making the real-data calibration gain visible across the nine acquisition-disjoint files.

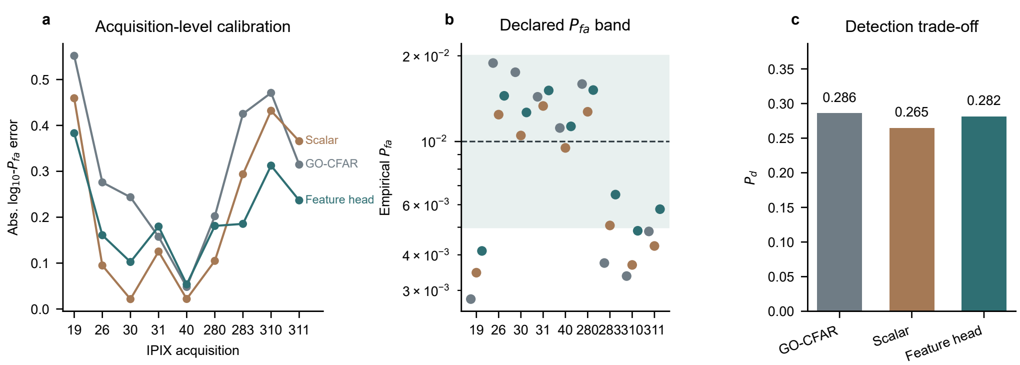


Figure 3: Retrospective IPIX acquisition-level result. Frozen calibration artifacts are applied to nine acquisition-disjoint measured sea-clutter files, with performance summarized by acquisition-level false-alarm calibration and primary-bin detection probability.

The retrospective result is important because all evaluated acquisitions are scored with frozen feature-head and scalar calibration artifacts. Under that protocol, scalar calibration improved over GO-CFAR, and the feature head gave the lowest acquisition-level log error. This ordering matches the design of BC-DRCFAR. A scalar correction shifts the threshold globally; the feature head responds to the background descriptors of each acquisition. The result is a measured-data trade-off with stronger false-alarm calibration and P_{d} close to GO-CFAR.

The IPIX result adds the measured-data test beyond the controlled synthetic benchmark. The acquisitions differ in their empirical clutter statistics, target range bins, and polarization behaviour, but they are evaluated with the same declared operating point and frozen calibration artifacts. The feature head keeps the macro false-alarm probability nearly nominal and improves the acquisition-level log error over GO-CFAR. The small change in primary P_{d} is also informative: the reliability gain is achieved mainly through false-alarm calibration while the target-separation behaviour remains close to the GO-CFAR reference. Table 3 gives the measured-data calibration trade-off across the development and retrospective settings.

Table 3: Retrospective IPIX calibration trade-off. The feature head is the main external-acquisition reliability result, while scalar and low-rank variants are mechanistic comparisons.

Family	Macro P_{fa}	Factor-2 metric	Primary P_{d}	Abs. log10 error
Development scalar	0.009792	0.845	0.375	0.278
Development low-rank	0.011479	0.200	0.393	–
Retrospective scalar	0.008332	0.830	0.265	0.213
Retrospective feature head	0.009998	0.846	0.282	0.199

5.3. Ablations: background descriptors drive the calibration gain

Feature ablation shows that the full background-conditioned head is the most stable deployable choice. Removing anchor features or polarization information degrades the external trade-off, whereas removing uncertainty or series-shift features has a smaller effect. Anchor features provide the strongest reference for the operating-point scale, and polarization information adds acquisition context beyond a global scalar correction. The ablation pattern ties the gain to structured background descriptors instead of generic model capacity.

The ablation results also clarify how the background is used. Anchor features provide a stable reference for the nominal false-alarm scale. Polarization features carry acquisition context that changes the distribution of clutter returns and the way high-amplitude reference cells should be interpreted. Uncertainty and series-shift descriptors refine this context, but their removal has a smaller effect in the evaluated setting. The full feature head therefore combines a stable operating-point anchor with acquisition-specific background information.

This ablation pattern supports the central design choice. The full head is not explained by a single decorative feature group. Anchor and polarization features carry the strongest external-acquisition signal, while the auxiliary descriptors preserve a compact description of uncertainty and series displacement. The background-conditioned family remains close across several ablations, showing that the gain is distributed across calibration features and remains stable across multiple descriptor choices.

5.4. Temporal and external audits: reliability remains traceable

The long-horizon holdout shows how calibration behaves under temporal drift. Scalar calibration changed from early $P_{\mathrm{d}} = 0.4663$ to late $P_{\mathrm{d}} = 0.2625$, while the feature head changed from 0.4742 to 0.2782. Under the same split, the feature head reduced the worsening in calibration error. Fold-level audits identified real source difficulty in the hardest folds, beyond sample-count imbalance. Domain audits placed IPIX inside the learned acceptance region, and the external-domain checks provided a reproducible transfer evaluation beyond the main IPIX setting.

These audits make the reliability result traceable. The long-horizon split shows that temporal drift affects both detection and calibration in measured sea clutter. The feature head reduces the calibration worsening under the same temporal split, matching the behaviour targeted by background conditioning. The fold-level audit links difficult folds to source structure, and the external-domain checks document how far the current calibration evidence can be followed beyond the main IPIX evaluation. Table 4 summarizes the ablation and reliability checks that support this mechanism.

Figure 4 provides the mechanism-level counterpart to Table 4. It links feature ablation, long-horizon drift, source-fold structure, and external-domain transfer into one reliability chain, showing that the calibration gain is tied to background information rather than to a single split or a single reporting metric.

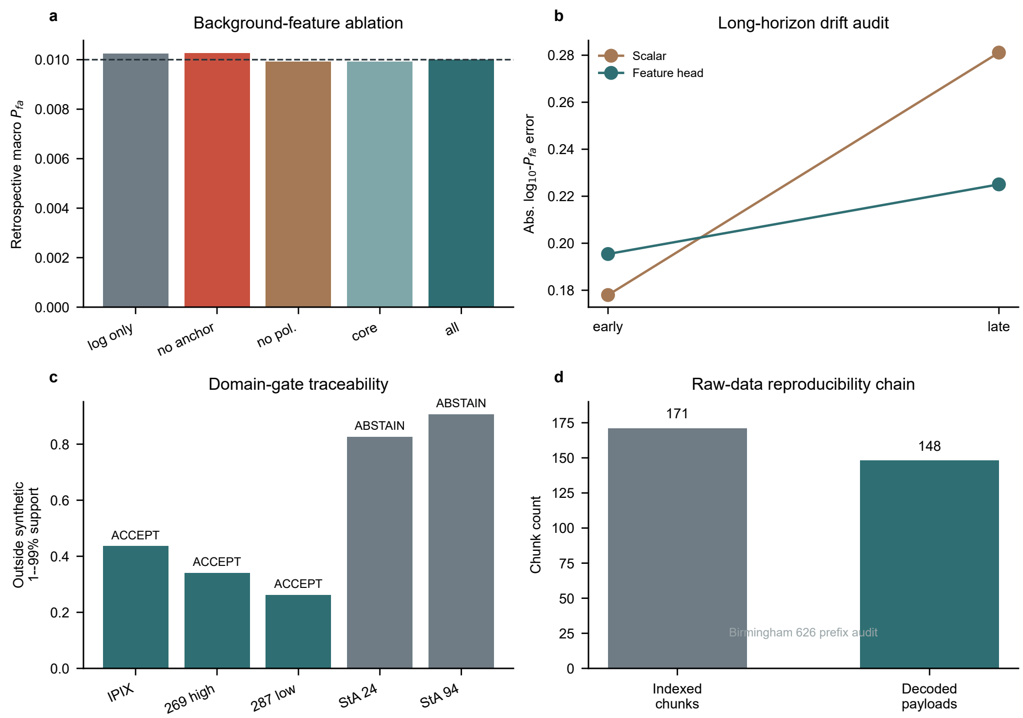


Figure 4: Mechanism and reliability audit. Background-feature ablations are presented together with temporal, source, and external-domain checks to show how acquisition-conditioned calibration produces the observed false-alarm reliability gain.

Table 4: Ablation and reliability audit summary. The table combines mechanism checks from feature ablation with temporal, source, and external-domain reliability audits.

Check	Result	Interpretation
Feature ablation	Removing anchor features or polarization information changes the external trade-off more than removing auxiliary uncertainty or series-shift terms	Anchor and polarization descriptors carry the strongest calibration context
Background-conditioned family	Several ablated feature-head variants remain close to the full model on retrospective macro P_{fa} and P_{d}	The calibration gain is distributed across background features and remains stable across descriptor choices
Long-horizon IPIX holdout	Feature head softens calibration drift under temporal split	Temporal reliability audit
Fold 3/4 source audit	Difficult folds reflect genuine source difficulty	Source-difficulty audit
Birmingham 626	148 prefix chunks decoded from 171 indexed records	Raw-data reproducibility audit
St Andrews/domain audits	External-domain checks evaluate transfer reliability	Domain-reliability audit

6. Discussion

BC-DRCFAR advances CFAR detection by treating false-alarm calibration as a background-conditioned decision problem. In nonstationary sea clutter, the reference background is more than a local power estimate. Its distribution, temporal dependence, tail behaviour, and ordered reference structure all affect whether a declared P_{fa} remains reliable within an acquisition.

This framing changes the role of the reference window. In classical CFAR, the reference cells mainly supply the local clutter level used to scale a threshold.

In BC-DRCFAR, the same cells also describe the calibration condition. A heavy-tailed background, a shifted reference series, or a temporally persistent clutter state is treated as information about how the threshold should be set. That is the central methodological contribution: CFAR calibration is made conditional on the background actually seen by the detector.

This view separates BC-DRCFAR from the usual extensions of classical CFAR. CA-CFAR, GO-CFAR, SO-CFAR, and OS-CFAR retain a clear thresholding interpretation, but their reference-window summaries are exposed to heavy-tailed, heterogeneous, and mismatched clutter [6, 5, 13]. Adaptive and statistical CFAR methods address this problem through richer clutter models, robust reference processing, and multi-feature probability models [8, 16, 21, 22, 23]. BC-DRCFAR keeps the declared false-alarm probability as the operating language and makes the threshold calibration depend on the observed background. The contribution is a calibrated extension of CFAR to acquisition-specific sea-clutter structure.

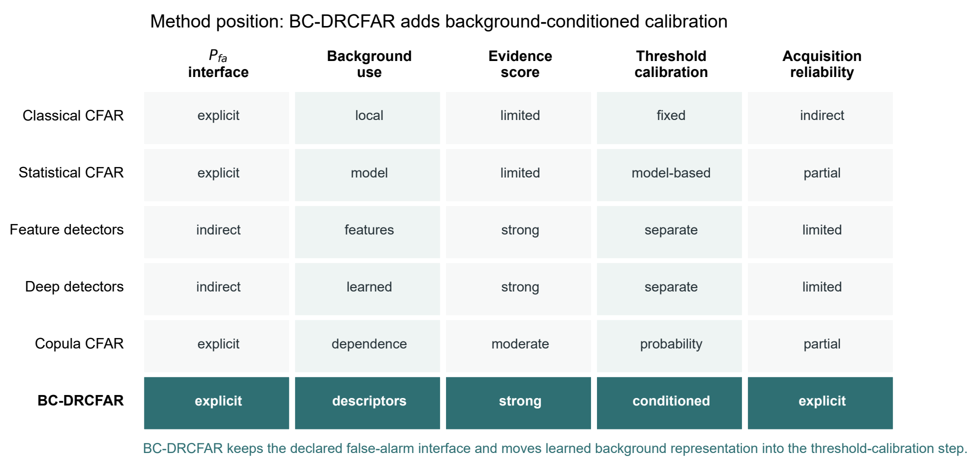
The comparison also explains why acquisition-level false-alarm summaries are central to this problem. Pooled P_{fa} gives the average operating point, while individual acquisitions reveal how consistently the detector holds that point under changing clutter. The factor-2 violation rate and log-scale false-alarm error used here make that behaviour visible. BC-DRCFAR is designed around those acquisition-level quantities, which is why its main advantage appears in reliability metrics as well as in the pooled rate.

The comparison with feature-based and learning-based detectors is equally important. Multi-domain feature fusion, reconstruction models, and manifold or geometric detectors mainly improve target–clutter separability in complex maritime scenes [20, 31, 33]. BC-DRCFAR instead puts the operating point at the centre of the model. Its score branch orders target evidence, while the feature head uses background descriptors to set the threshold for the declared P_{fa}. This division explains the retrospective IPIX result: the main gain appears in acquisition-level false-alarm calibration, with the primary detection probability remaining close to GO-CFAR.

This division of labour is useful for radar deployment. Operators choose a false-alarm operating point because it has operational meaning. BC-DRCFAR keeps the score and calibration tasks connected but distinct. The score branch captures target evidence; the feature head translates the declared operating point into the threshold required by the observed clutter.

Against the comparison set of recent sea-clutter detectors, the main innovation is therefore the placement of learned background representation inside the CFAR calibration step. Feature-fusion and reconstruction detectors improve the evidence score; copula and statistical CFAR methods improve the probability model; classical CFAR variants improve reference-window robustness. BC-DRCFAR adds an acquisition-conditioned map from declared P_{fa} to the decision threshold. That map is what lets the method preserve the CFAR operating language while responding to measured background structure [20, 28, 22, 23, 32].

Figure 5 makes this positioning explicit. The comparison matrix separates the operating interface, background representation, evidence modelling, threshold calibration, and acquisition-level reliability roles: classical and statistical CFAR methods strengthen the reference-window or probability model, feature and reconstruction detectors strengthen the evidence score, and BC-DRCFAR inserts learned background representation into the threshold-calibration step that controls the declared operating point.



	P_{fa} interface	Background use	Evidence score	Threshold calibration	Acquisition reliability
Classical CFAR	explicit	local	limited	fixed	indirect
Statistical CFAR	explicit	model	limited	model-based	partial
Feature detectors	indirect	features	strong	separate	limited
Deep detectors	indirect	learned	strong	separate	limited
Copula CFAR	explicit	dependence	moderate	probability	partial
BC-DRCFAR	**explicit**	**descriptors**	**strong**	**conditioned**	**explicit**

Figure 5: Position of BC-DRCFAR relative to recent sea-clutter detection methods. The comparison matrix highlights the central contribution of the method: learned background representation is inserted into the CFAR threshold-calibration step while the declared false-alarm operating point remains the decision interface.

The experiments support this mechanism across controlled and measured settings. On the synthetic benchmark, BC-DRCFAR reduces factor-2 false-alarm violations and log-P_{fa} error while improving P_{d} at 0 dB. On retrospective IPIX acquisitions, frozen feature-head calibration keeps macro P_{fa} near the nominal value and improves acquisition-level error over scalar calibration and GO-CFAR. The ablations point to the same conclusion: anchor features and polarization information carry calibration context that a background-conditioned threshold head can use directly.

The reliability audits add the deployment perspective. The long-horizon split shows that temporal drift changes both detection probability and calibration error, while background conditioning softens the calibration drift. Fold-level audits show that the hardest folds reflect genuine source difficulty and sample-count structure. Birmingham 626 adds a raw-data reproducibility check: the remote ZIP, MAT/HDF5 schema, B-tree chunk index, and chunk payloads can be traced to decoded complex IQ blocks. That traceable path extends the evidence chain for acquisition-level CFAR calibration.

Taken together, the experiments support a clear contribution to radar signal processing: background-conditioned calibration turns CFAR from a local thresholding recipe into an acquisition-aware reliability mechanism. The method retains the declared false-alarm interface that makes CFAR practical, and adds a data-driven threshold calibration rule tied to the clutter state. This is the step that connects modern feature representation with the operational discipline of false-alarm control.

This framing also clarifies the role of the reliability audits. The audits are part of the same evidence chain. The synthetic benchmark verifies the mechanism under controlled clutter shifts. The retrospective IPIX study tests frozen calibration on measured sea clutter. The ablations identify which background information drives the calibration behaviour. The temporal and source audits show how the same decision rule behaves when acquisition conditions change. Together these experiments support BC-DRCFAR as a calibrated radar signal-processing method whose reliability evidence is tied to its calibration mechanism.

7. Conclusion

BC-DRCFAR addresses acquisition-level false-alarm instability by conditioning CFAR calibration on the observed clutter background. It keeps the declared P_{fa} as the operating point, then uses background descriptors and a feature-head threshold to adapt the decision rule to nonstationary sea clutter. The scale-equivariant score branch preserves target-evidence ordering while the calibration branch sets the acquisition-specific threshold.

Controlled and measured evaluations give the same conclusion. On the synthetic benchmark, BC-DRCFAR reduces the factor-2 violation rate from

83.3% to 18.8%, lowers the mean absolute log10-P_{fa} error from 1.070 to 0.174, and raises P_{d} at 0 dB from 0.740 to 0.835. On retrospective IPIX acquisitions, the feature head keeps the macro false-alarm probability near the nominal level and satisfies the acquisition factor-2 condition in 7 of 9 acquisitions. The ablation and reliability audits point to the same mechanism: background-conditioned calibration gives CFAR a practical route to reliable radar target detection in nonstationary sea clutter.

CRediT authorship contribution statement

Zixuan Liu: conceptualization, methodology, software, formal analysis, investigation, visualization, writing - original draft. Wei Xiong: conceptualization, validation, supervision, writing - review and editing, project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The manuscript uses public radar data sources and project-generated synthetic benchmarks. The IPIX radar data are available from the McMaster University IPIX archive, and the Birmingham 626 raw-data audit uses publicly released maritime radar data. The source code, processed evaluation summaries, synthetic benchmark configuration, figure-generation materials, and manuscript source files supporting the reported figures and tables are archived in Zenodo at `https://doi.org/10.5281/zenodo.21855949`. The corresponding GitHub repository is available at `https://github.com/niubisile-wq/bcdrcfar-dsp`.

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